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Train Ticket Booking system using Linked List
#include <stdio.h>



#include <stdlib.h>
struct Node

{

 int ticket;

 struct Node *next;

};
struct Node *confirmed=NULL;

struct Node *waiting=NULL;
int seats = 4;

int fare = 500;
void add(struct Node **head,int ticket)

{

 struct Node *newNode, *temp;

 newNode=(struct Node*)malloc
(sizeof(struct Node));
newNode->ticket=ticket;

 newNode->next=NULL;

if (*head==NULL)
*head=newNode;

 else

 {

 temp=*head;

 while (temp->next!=NULL)

 temp=temp->next;
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```
 temp->next=newNode;
 }
}
void book(int ticket)
{
 if (seats>0)
 {
 add(&confirmed,ticket);
 seats--;
 printf("Ticket %d confirmed.\n",ticket);
 printf("Fare =Rs.%d\n",fare);
 }
else
 {
 add(&waiting,ticket);
 printf("No seat available.\n");
 printf("Ticket %d added to waiting list.\n",ticket); }
}
void cancel(int ticket)
{
struct Node *temp=confirmed;
 struct Node *prev=NULL;
while(temp!=NULL)
```

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{
if (temp->ticket==ticket)
break;
prev=temp;
temp=temp->next;
}
if(temp==NULL)
{
printf("Ticket not Found.\n");
return;
}
if(prev==NULL)
confirmed=temp->next;
else
prev->next=temp->next;

free(temp);
printf("Ticket %d cancelled.\n",ticket);
printf("Refund=Rs.%d\n",fare);
if(waiting!=NULL)
{
temp=waiting;
waiting=waiting->next;
```

```
 add(&confirmed,temp->ticket);

 printf("Ticket %d is now confirmed.\n",temp->ticket);
free(temp);

 }

 else

 {

 seats++;

 }

}

void display()

{

 struct Node *temp;

   printf("\nConfirmed:");

    temp=confirmed;

   while(temp!=NULL)

    {

    printf("%d ",temp->ticket);

    temp=temp->next;

    }

   printf("\nWaiting:");

    temp=waiting;


   while(temp!=NULL)
```

```
  {
   printf("%d ",temp->ticket);
   temp=temp->next;
   }
  printf("\n");
}
int main()
{
 int choice,ticket;
  while(1)
   {
   printf("\n1.Book Ticket");
   printf("\n2.Cancel Ticket ");
   printf("\n3.Display");
   printf("\n4.Exit");
   printf("\nEnter choice:");
      scanf("%d",&choice);
  if(choice==1)
   {
   printf("Enter ticket number:");
   scanf("%d",&ticket);
```

```
book(ticket);
}
else if(choice==2)

{
printf("Enter ticket number:");
scanf("%d",&ticket);
cancel(ticket);
}
else if(choice==3)
{
display();
}
else if(choice==4)
{
break;
}
else
{
printf("Invalid choice.\n");
}
}
```

```
return 0;

}
```

\\OUTPUT\\

```
1.Book Ticket

2.Cancel Ticket

3.Display
4.Exit

Enter choice:1

Enter ticket number:1

Ticket 1 confirmed.

Fare =Rs.500

1.Book Ticket

2.Cancel Ticket

3.Display

4.Exit

Enter choice:1

Enter ticket number:2

Ticket 2 confirmed.

Fare =Rs.500

1.Book Ticket

2.Cancel Ticket
```

3.Display

4.Exit

Enter choice:1

Enter ticket number:3

Ticket 3 confirmed.

Fare =Rs.500

1.Book Ticket

2.Cancel Ticket

3.Display

4.Exit

Enter choice:1
Enter ticket number:4

Ticket 4 confirmed.

Fare =Rs.500

1.Book Ticket

2.Cancel Ticket

3.Display

4.Exit

Enter choice:1

Enter ticket number:5

No seat available.

Ticket 5 added to waiting list.

1.Book Ticket

2.Cancel Ticket

3.Display

4.Exit

Enter choice:1

Enter ticket number:6

No seat available.

Ticket 6 added to waiting list.

1.Book Ticket

2.Cancel Ticket

3.Display

4.Exit

Enter choice:3

Confirmed:1 2 3 4
Waiting:5 6

1.Book Ticket

2.Cancel Ticket

3.Display

4.Exit

Enter choice:2

Enter ticket number:3

Ticket 3 cancelled.
Refund=Rs.500

Ticket 5 is now confirmed.

1.Book Ticket

2.Cancel Ticket

3.Display

4.Exit

Enter choice:3

Confirmed:1 2 4
5

Waiting:6